Multi-Material 3D-Printed Tensegrity Energy Absorbers for Planetary Landing and CubeSat Payload Protection

Utah NASA Space Grant Consortium Fellowship Proposal, 2026–2027

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1 Introduction and Motivation

I am applying for a Utah NASA Space Grant Consortium fellowship to support a year of focused research on **multi-material 3D-printed tensegrity-inspired structures for spacecraft and payload energy absorption**. Every NASA mission that lands on a planet, returns a sample to Earth, or deploys a CubeSat from a launch vehicle has to dissipate a large pulse of kinetic energy in a very short time. Mars Pathfinder and the Mars Exploration Rovers landed on inflatable airbags qualified through 52 vacuum-chamber drop tests at velocities up to ∼25 m/s [Adams, 2004]; later crashworthy planetary landers have used composite-honeycomb energy absorbers designed to a ∼20 psi crush stress and a 20 g floor-acceleration limit [Jackson et al., 2014]; CubeSats and instruments routinely sit on foam or elastomer isolators to survive launch shock and vibration [NASA Goddard Space Flight Center, 2019]. These incumbent attenuators work, but they are heavy, single-use, and tuned to a narrow load profile—a design space first formalized by Cloutier [1966] as a trade among crushing stress, density, and stroke.

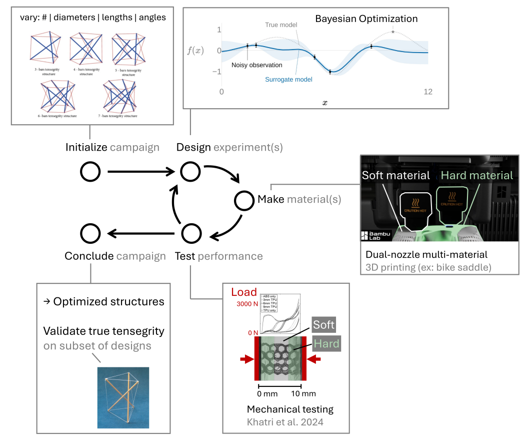

Figure 1: Closed-loop print–test–optimize workflow I will run on the Bambu Lab H2D: parameterized PLA + TPU tensegrity-inspired specimens are fabricated, characterized under quasi-static compression and drop-weight impact, and fed back to a Bayesian optimizer that recommends the next geometries to print.

Tensegrity-inspired structures—networks of rigid struts suspended in a continuous web of tension elements—offer a programmable alternative. NASA Ames' NIAC-funded *Super Ball Bot* project demonstrated that tensegrity geometries can survive impacts on the order of 15 m/s and combine landing, payload protection, and surface mobility in a single packable structure [Caluwaerts et al., 2014, Agogino et al., 2018, Sabelhaus et al., 2015, Vespignani et al., 2018], and the concept has been extended to ocean-world rideshare rovers [Deitrich et al., 2022]. What has been missing is a fast, low-cost way to design and *experimentally* validate tensegrity-inspired geometries against NASA-relevant impact scenarios. Recent work shows that tensegrity-inspired unit cells can be directly 3D-printed on consumer-grade FDM hardware and retain a load-limiting plateau under drop-weight impact [Pajunen et al., 2019]. Multi-material rigid–soft FDM (rigid struts + flexible tension skins) yields tunable energy absorption without delamination [Ye et al., 2023, Khatri and Egan, 2024], and our group has access to a dual-nozzle **Bambu Lab H2D** printer that makes this multi-material workflow practical for me to run as a single-student campaign.

I propose to use that capability to run a focused, year-long **print–test–optimize** campaign aimed squarely at two NASA-relevant load cases: (1) low-velocity *lander-class* crush attenuation and (2) launch-/deployment-class *shock isolation* for small-payload (CubeSat-class) hardware. This work directly supports the Utah NASA Space Grant Consortium's mission of preparing students for the

NASA workforce and aligns with two STMD priorities visible across the Ames NIAC tensegrity line and the In-Space Manufacturing (ISM) / FabLab roadmap: lightweight load-limiting structures and multi-material additive manufacturing [National Aeronautics and Space Administration, 2022, Prater et al., 2019, Agogino et al., 2018].

2 Background and Relation to NASA Priorities

Tensegrity in space systems. Tensegrity structures [Skelton and de Oliveira, 2009, Sultan, 2009] have a long history in NASA-funded research. Caluwaerts, Agogino, SunSpiral, and collaborators at NASA Ames proposed tensegrity-based planetary probes that combine landing and surface mobility in a single deployable structure [Caluwaerts et al., 2014, Agogino et al., 2018, Sabelhaus et al., 2015, Vespignani et al., 2018], and the line continues into Titan-class ocean-world rideshare concepts [Deitrich et al., 2022]. Pajunen et al. [Pajunen et al., 2019] showed that 3D-printable *tensegrity-inspired* unit cells exhibit a long plateau in their force–displacement response under drop-weight impact—exactly the behavior desired in a crush core, where peak transmitted force must stay below a hardware/biological tolerance. Compared to honeycomb or foam, tensegrity geometries can be *tuned* (struts, cables, topology) without changing manufacturing process, opening the door to mission-specific designs.

Multi-material 3D printing as an enabling technology. In-space manufacturing is an active NASA priority: the ISS has hosted FDM printers since 2014, and on-demand fabrication is expected to be central to long-duration missions [Prater et al., 2019]. Demonstrating that tensegrity energy-absorbers can be reliably produced on a single FDM platform with two materials (PLA struts + TPU tension elements) contributes to that broader trajectory while staying achievable within a single fellowship year. The **Bambu Lab H2D**'s dual-nozzle architecture and Automatic Material System let me switch between rigid PLA and flexible TPU within a single print, dramatically reducing iteration time on each new geometry. I expect to print on the order of **50–100 specimens** over the year.

Bayesian optimization of impact-absorbers. The design space (strut diameter and length, TPU skin thickness, tiling and topology) is too large to sweep exhaustively, and physical specimens are expensive in time, not just filament. Bayesian optimization (BO) [Shahriari et al., 2016, Frazier, 2018, Zhang et al., 2021] fits the problem: each printed specimen updates a Gaussian process surrogate that recommends the next geometry to print and test. BO has been applied to architected and metamaterial design [Mo et al., 2023, Zhang et al., 2021] and to closed-loop materials discovery in domains ranging from concrete [Ament et al., 2023] to organic emitters [Strieth-Kalthoff et al., 2024]; here I apply it directly to a NASA-relevant impact-attenuation problem with fully physical drop-weight tests, which the BO-in-AM literature identifies as an open gap [Zhang et al., 2021].

3 Research Objectives

I have three concrete, measurable objectives for the fellowship year:

1. **Build a parameterized tensegrity-inspired unit-cell family** that prints reliably on the Bambu Lab H2D in PLA + TPU, with verified geometric fidelity (CT or caliper-and-mass) on at least 5 baseline geometries.
2. **Quantify NASA-relevant impact response** for ≥ 30 distinct geometries via quasi-static compression and drop-weight tests, extracting peak transmitted force, specific energy absorption (SEA), and crush-plateau efficiency. Compare against a 3D-printed honeycomb baseline of equal areal density.
3. **Close the loop with Bayesian optimization** to find geometries that minimize peak transmitted force at a target SEA for two NASA-motivated load cases: a low-velocity "lander" drop ($\sim$3–6 m/s, 20–50 J) and a high-frequency "CubeSat" shock pulse [NASA Goddard Space Flight Center, 2019].

4 Proposed Approach and Methods

Design. I will parameterize a family of tensegrity-inspired unit cells in Fusion 360—strut diameter and length, TPU skin thickness, connectivity topology, and tiling pattern—following the wrapping-based core/skin strategy of Ye et al. [Ye et al., 2023] so that PLA struts remain mechanically interlocked with continuous TPU tension members and do not delaminate under cyclic loading [Khatri and Egan, 2024].

Fabrication. All specimens will be printed on the **Bambu Lab H2D** in our lab, using its dual-nozzle, multi-material capability to produce PLA + TPU parts in a single build. I will keep a **print log** (slicer settings, AMS routing, post-print mass and geometric checks) so that downstream test data can be traced back to a specific build configuration. Early in the year I will validate PLA–TPU interface robustness on a reduced set of test coupons before committing to full unit-cell prints.

Testing. Two complementary protocols, both NASA-relevant:

- *Quasi-static compression* on a benchtop universal testing machine to extract force–displacement curves, plateau stress, and densification strain (analogous to the characterization done for honeycomb crush cores in EDL design).
- *Drop-weight impact* from instrumented fixed-mass drops to measure peak transmitted force and SEA at lander-relevant velocities, following the protocol of Pajunen et al. [Pajunen et al., 2019]. A piezoelectric accelerometer or load cell at the base captures the transmitted pulse.

A 3D-printed aluminum-equivalent honeycomb baseline of equal areal density will be tested under the same conditions to anchor performance claims to a NASA incumbent.

Bayesian optimization loop. I will implement a BO loop in Python using BoTorch / Ax [Balandat et al., 2020] that operates directly on experimental measurements. Each round of physical tests updates a Gaussian process surrogate; the acquisition function (expected hypervolume improvement for the multi-objective peak-force / SEA trade-off) recommends the next batch of ∼3–5 designs to print and test. This closed loop is the experimental analogue of the multifidelity BO framework of Mo et al. [Mo et al., 2023], but here grounded entirely in physical data so that conclusions are not limited by simulation fidelity—an important property for NASA-relevant impact problems where rate-dependent TPU damping is hard to model accurately.

Quarterly Task Plan

Task	Q1 (Fa)	Q2 (Wi)	Q3 (Sp)	Q4 (Su)
Literature review and NASA load-case scoping	•			
Bambu H2D PLA/TPU print process validation	•	•		
Unit-cell parameterization and CAD	•	•		
Quasi-static compression campaigns		•	•	•
Drop-weight impact testing			•	•
BO-guided design iteration			•	•
UNSGC annual conference presentation			•	
Final report and manuscript preparation				•

Table 1: Quarterly task plan for the 2026–2027 fellowship year. Q1 = Fall 2026, Q2 = Winter 2027, Q3 = Spring 2027, Q4 = Summer 2027.

5 Expected Outcomes and Broader Impact

By the end of the fellowship year I expect to deliver: (i) a publicly released data set of ∼50–100 PLA/TPU tensegrity-inspired specimens with quasi-static and drop-weight impact characterization, (ii) an open-source Python implementation of the experimental BO loop, and (iii) a poster or talk at the annual UNSGC conference (a deliverable required of all Space Grant fellows) and a draft journal manuscript targeting *Smart Materials and Structures* or the *ASME Journal of Mechanical Design*.

The broader impact is twofold. Scientifically, this is an early empirical demonstration that a single student, with one multi-material FDM printer and a Bayesian optimization loop, can systematically discover impact attenuators that out-perform a NASA-incumbent geometry on a relevant load case. The resulting dataset and methodology are designed to seed follow-on proposals to NASA NIAC (in the Super Ball Bot lineage [Agogino et al., 2018, Deitrich et al., 2022]), STMD/SBIR ISM calls, and NASA EPSCoR, with a clear materials-qualification pathway from PLA/TPU discovery specimens toward space-qualified polymers (PEI/PEKK/PEEK) screened under MISSE-style protocols [Finckenor and McElderry, 2023]. Educationally, this fellowship would give me hands-on training in exactly the skills NASA's strategic plan calls out for the future workforce [National Aeronautics and Space Administration, 2022]: multi-material additive manufacturing, mechanical testing, and data-driven design. I plan to use the experience as a springboard into graduate study in aerospace structures or in-space manufacturing.

6 Qualifications and Mentoring Plan

My background. [TODO: Marcus to fill in: current year (e.g., sophomore/junior), GPA, relevant coursework (mechanics of materials, machine design, programming), prior 3D printing or research experience, and any space-related coursework or extracurriculars (e.g., AIAA, BYU CubeSat involvement).]

Mentoring structure. I will work under Prof. Sterling G. Baird in the Department of Mechanical Engineering, with weekly one-on-one meetings (30 min) for technical guidance and professional development, and weekly lab group meetings (1 hr) covering progress updates, literature, and skill workshops. Prof. Baird's master's student will provide day-to-day co-mentoring on the Bambu H2D, mechanical testing equipment, and the Python BO codebase. I will keep a public lab notebook (GitHub) and present progress monthly to the lab. The PI will provide the required 50% non-federal cost share from laboratory startup funds.

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